SANKET AI (■■■■■ AI)

Enterprise Multimodal Brand Governance, Automated Inspection & Generative Remediation SaaS

ROUND 3 — FINAL PRODUCT SUBMISSION

● 100% Shipped & Certified

Team: Dark Lord • **Lead:** Suresh Savant

Live Project Link:	https://sanket-ai.onrender.com (Zero-auth public access)	**Automated Test Suite:**	**48 / 48 Passed (100% Pass Rate)**
GitHub Repository:	https://github.com/sureshsavant576-create/dark-lord (Public)	**Evaluator Presets:**	Nike Certified • Off-Brand Auto-Fix • GRB Heritage

1. Executive Summary & Real-World Fit

Modern marketing and creative teams release hundreds of campaign assets across social, display, and web every week. A single off-brand creative with non-compliant colors, forbidden legal claims, or jarring aesthetic tone inflicts immediate damage to enterprise equity. Traditional LLM-only review tools fail because models hallucinate exact HEX color codes and miss strict policy boundaries. **Sanket AI** solves this via an industrial-grade **Two-Stage Hybrid Architecture**: deterministic CIELAB ΔE mathematical color science and zero-hallucination regex token scanners execute first, followed by Google Gemini 2.5 Flash Vision for contextual multimodal aesthetic synthesis.

2. The Three Committed Improvements Delivered in Round 3

In Round 2, Team Dark Lord committed to delivering exactly three major capabilities in Round 3. All three have been fully built, unit-tested, and verified in the live product:

#	Improvement Name & Description	Delivered Capability & Architectural Proof
1	**Interactive Brand Co-Pilot (Remediation Chatbot)** *Submitted Commitment:* Add an AI brand assistant next to the compliance report. Creators can chat directly with the brand's AI persona to get suggested copy rewrites, fix tone issues, and receive step-by-step guidance.	• Built BrandCopilotService and endpoint POST /inspect/{brand_id}/copilot-chat. • Dynamically adopts the brand's exact tone, values, avoidances, and mission. • Generates **3 one-click compliant copy rewrites** that automatically strip banned terms. • Embedded directly in Check Vibe with 1-click 'Apply to Creative' copy button.
2	**One-Click Generative Auto-Remediation (Auto-Fix)** *Submitted Commitment:* Move from just detecting errors to automatically fixing them. Offer an 'Auto-Fix' button that automatically shifts off-brand colors to the closest approved brand palette centroid.	• Built CreativeRemediationService and endpoint POST /inspect/{brand_id}/auto-remediate. • Converts raster image into CIELAB Lab space and identifies color clusters exceeding $\Delta E \geq 12.0$ threshold. • Programmatically remaps off-brand pixels to approved brand centroids while preserving local luminance, shadows, and textures. • Produces ready-to-deploy certified graphic ($\Delta E < 5.0$, 95%+ score) with side-by-side comparison modal and download link.

3	**Instant Drag-and-Drop Pre-Flight Scanner** *Submitted Commitment:* Add an instant client-side scanner that runs the moment an image is dragged into the browser—before clicking submit. Immediately show color temperatures, negative space ratios, and predicted visual vibe.	• Built sub-millisecond client-side HTML5 Canvas matrix engine paired with POST /inspect/preflight-scan. • Computes **Kelvin Color Temperature** (e.g., Warm 3000K, Neutral 5400K, Cool 6500K). • Computes **Whitespace & Negative Space Density** ratio using Laplacian local variance. • Predicts Visual Vibe (Minimalist, Bold, Cyberpunk, Warm Heritage) and previews top 5 swatches in **< 35ms**.

3. Technical Architecture & Two-Stage Inspection Pipeline

Layer	Technology	Implementation Purpose & Deterministic Guarantees
Backend Core	Python 3.15, FastAPI, Uvicorn, Pydantic v2	Asynchronous ASGI server with strict schema validation, structured error recovery, and zero hanging requests under high burst load.
Color Science	CIELAB, CIE76 Euclidean ΔE, Pillow, NumPy	Quantizes image pixels into Lab color space. Enforces strict mathematical threshold (ΔE < 15.0). Prevents off-brand color hallucination.
Lexical Scanner	Python Regex Engine (`re`), Tokenizer	Zero-hallucination regex token matching against brand avoidances and prohibited claims. Drops deterministic compliance score to 0.0% on hard violation.
Multimodal AI	Google Gemini 2.5 Flash / Flash-Lite API	Asynchronous httpx client evaluating subjective brand alignment: visual aesthetic resonance, humor vs. seriousness, and typography harmony.
Frontend UI/UX	Vanilla ES6+, HTML5 Canvas, Modern CSS3	Fast, lightweight, mobile-responsive UI with SVG icon system, zero external heavy bloated JS frameworks, client-side eyedropper loupe, and drawer navigation.
Persistence	JSON Database with Thread-Safe Atomic Writes	Full CRUD lifecycle: Brand Registration, Logo Asset Staging, Historical Audit Logs, and Cascade Deletion without database lock corruption.

4. Complete List of Shipped Features (Full Product Suite)

- **Company Portal Brand Onboarding:** Full corporate identity registration (tone, voice, personality, visual style, motto, approved HEX swatches, and prohibited claim avoidances).
- **Automated Reference Feature Extraction:** Upload posters/brand manuals to auto-extract dominant palette centroids, whitespace density, and complexity indices.
- **Interactive Colour Intelligence Studio:** Client-side HTML5 Canvas eyedropper with magnifying pixel loupe for precision color selection.
- **Instant Drag-and-Drop Pre-Flight Scanner (Round 3):** Sub-35ms client & server diagnostics for Kelvin temperature, whitespace ratio, and predicted visual vibe.
- **Deterministic Policy Lexicon Scanner:** Regex token matching enforcing hard zero-tolerance penalties for false promises, illegal sales claims, and banned terminology.
- **CIELAB ΔE Perceptual Color Matching:** Mathematical Euclidean distance calculation in perceptually uniform Lab space (< 15.0 delta threshold).
- **Google Gemini 2.5 Flash Multimodal Reasoning:** High-speed subjective aesthetic synthesis with structured JSON schemas and graceful fallback resilience.

- **Interactive Brand Co-Pilot (Round 3):** Conversational AI brand guardian offering 3 one-click compliant copy rewrites and real-time guidance.
- **1-Click Generative Auto-Remediation (Round 3):** Programmatically shifts off-brand pixels to approved brand centroids with side-by-side comparison and download.
- **Evaluator & Judge 1-Click Test Presets:** Zero-setup testing presets (Nike Certified Compliant, Off-Brand Auto-Fix Demo, and GRB Heritage Traditional).
- **Company Account Management Dashboard:** Multi-company management, live profile editing, and permanent audit history with cascade deletion safeguards.
- **Responsive Dashboard UI:** Multi-page architecture for Check Vibe, Company Portal, and Account with sleek drawer pull tab and mobile compatibility.
- **Automated Security Audit & Hardening:** Strict CSP headers, MIME verification, path traversal prevention, and rate-limiting safeguards.

5. Reliability Under Normal Use ('Hold Up Without You')

A critical requirement for Round 3 is that the application must hold up without team intervention: bad inputs, empty states, page refreshes, and rapid repeated actions must never crash or hang the interface. We engineered four layers of resilience:

- **Graceful Fallback on API Failures:** If external AI quotas are throttled or API keys are missing, the system automatically falls back to deterministic rule-based generators. The UI never freezes or shows raw stack traces.
- **Empty State & Bad Input Handling:** Forms feature HTML5 novalidate with custom JavaScript validation boundaries. Uploading non-image files, empty copy, or corrupt formats triggers clear, human-friendly guidance toasts without page reloads.
- **Idempotency & Double-Click Protection:** All audit and upload buttons immediately enter an animated loading state with disabled=true to prevent race conditions or duplicate database entries.
- **1-Click Judge Presets:** Judges can evaluate the full pipeline in 5 seconds by clicking the top presets without needing to source local test images.

6. Automated Testing Details & Verification Matrix

The codebase includes a comprehensive automated test suite powered by pytest and pytest-asyncio. **All 48 tests pass with a 100% success rate:**

Test Module	Tests	Scope & Coverage Description	Status
test_round3_improvements.py	4	Brand Co-Pilot chat, Pre-flight matrix diagnostics, Auto-remediation pixel shifts, and Round 3 REST endpoints.	**PASSED**
test_inspection_engine.py	5	End-to-end multimodal inspection, score weighting, and dual-engine fallback reconciliation.	**PASSED**
test_preprocessing.py	5	CIELAB color extraction, Delta-E math, whitespace ratio, and thumbnail downsampling.	**PASSED**
test_security_audit.py	11	MIME spoofing defenses, path traversal, file upload limits, and security headers.	**PASSED**
test_deletion_lifecycle.py	3	Safe cascade deletion, disk file cleanup, and orphan reference prevention.	**PASSED**
test_brand_identity.py	2	Brand standard creation, schema validation, and persistent storage integrity.	**PASSED**
test_api_endpoints.py	3	REST routing, HTTP response status codes, and error payloads.	**PASSED**
test_phase8_validation.py	8	Full-stack integration, multimodal joint inputs, and edge-case validation.	**PASSED**

test_ai_gateway.py	4	Google Gemini provider initialization, prompt construction, and provider failover.	**PASSED**
test_config.py	3	Environment variable parsing, directory auto-initialization, and settings validation.	**PASSED**
TOTAL	**48**	**100% Comprehensive Suite Pass Rate across all 48 test cases**	**48 / 48 PASS**

7. Product Roadmap & Future Development Plan

Phase 4: Video Stream Creative Inspection: Expand the multimodal inspection pipeline to evaluate video advertising (MP4, WebM) via automated keyframe extraction, scene-by-scene audio-visual lexicon scanning, and logo watermark tracking.

Phase 5: Multi-Hundred-Page Brand Manual PDF Chunking: Implement semantic RAG chunking and vector embeddings for massive 300+ page enterprise brand guideline documents containing dense design matrices and complex tabular policies.

Phase 6: Ad Platform Automated Webhook Enforcement: Connect direct OAuth integrations with Meta Ads Manager and Google Ads API to auto-flag, quarantine, or block non-compliant creative assets before paid media spend occurs.